uOttawa - Telfer
Workload Allocation Problem

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1 Description

1.1 Intro

The goal for this project is to create a course schedule. The schedule must incorporate professors, courses, time blocks, semesters, and classrooms. Meaning, each section (of the course) must be scheduled at a specific time (time block and semester combination) to a specific professor in a specific classroom.

In addition to creating a schedule, a web application will be developed with a functional back-end database and front-end user interface. This is done for the demonstration purposes and to make it easier to integrate with existing Telfer web structure.

1.2 Solution Approach

2 Optimization Model

2.1 Sets

P - professors - "part-timer" is added to set
C - courses
I - unique section ID
K - classrooms - "other" classroom is added to set
S - semesters
T - timeslots - "placeholder" is added for each TD
TD - times of day
DW - days of week
BL - blocks

2.2 Subsets

$CS_{cis,td}$ - possible course C section I per S, TD
$TTD_{t,td}$ - timeslots to times of day
$TDW_{t,dw}$ - timeslots to days of week
$TBL_{t,bl}$ - timeslots to blocks

2.3 Data Requirements

2.3.1 Time Data

tbw_{bl} - timeslots weights for block BL (for day of week equations)
$tc_{t,t}$ - timeslots concurrencies
$tb_{t,t}$ - timeslots that are back to back

2.3.2 Professor Data

pt_p - Required number of sections to be taught by professor
$pq_{p,c}$ - Professor to course qualifications
pcn_p - Indicates whether professor can teach at night or not
pcd_p - Indicates whether professor can teach outside of DMS or not
$pcp_{p,c}$ - Professor preference for a course
$ptp_{p,s,td}$ - Professor preference for a time
$pbp_{p,bl}$ - Professor preference for a block
ppp_p - Professor preference for a packed day

2.3.3 Classroom Data

kc_k - Classroom capacity

2.3.4 Course & Section Data

$csa_{c,i,s}$ - Course section attendance
$cct_{c,c,s}$ - These courses C must be scheduled together during S
$ccn_{c,c,s}$ - These courses C cannot be scheduled together during S
cp_c - Courses having part-timer penalty
cf_c - Courses having friday night penalty
$cb_{c,bl}$ - Course block preference

2.4 Decision Variables

2.4.1 Main Decision Variables

X_{pcistk} - assign a professor p to a section c, i, semester s, time t, room k.

A viable set of X, denoted XV, is defined by a following combination:

- where CS exists
- where pq exists
- where section attendance is between 80% and 100% of classroom capacity
- where 3 hour blocks with more than 75 students must be at night only
- where combinations of professors who can't teach at night and night blocks are removed
- where combinations of professors who can't teach outside of DMS and 'other' classrooms are removed

In the formulation this restriction should generally be denoted in the summation over X, however it is placed outside of the equation for better readability

$$X = \begin{cases} integer, & \text{if } p \in \{\text{part-timer}\} \text{ or } t \in \{\text{placeholder}\} \text{ or } k \in \{\text{other}\} \\ binary, & \text{otherwise} \end{cases}$$

2.4.2 Penalty and Reward Decision Variables

$TimeMax_s$ - Indicates a maximum of courses offered in each time of day, per semester s
$TimeMin_s$ - Indicates a minimum of courses offered in each time of day, per semester s

$DayMax_s$ - Indicates a maximum of courses offered in each day of week, per semester s
$DayMin_s$ - Indicates a minimum of courses offered in each day of week, per semester s

$KCount_k$ - Counts total sections assigned to each classrooms k
$CBLPen_{c,bl}$ - Counts the number of violations of block preference bl per course c
$CFNPen_c$ - Counts the number of violations of blocks scheduled to Friday night 3 hour block
$PDayT_{p,s,dw}$ - Indicates if a professor p is teaching during day of the week dw, semeseter s
$ProfBBV$ - Counts the number of back to back occurences for professor p

2.4.3 Hard Constraint Violation Indicators

$ConcI_{cst}$ - Counts the number of concurrencies of courses with itself
$ConcT_{c,cc,st}$ - Counts the number of concurrencies between courses where they should be together
$ConcN_{c,cc,st}$ - Counts the number of concurrencies between courses where they should not be together
$ProfTVp$ - Counts violations for required number of sections to be taught per professor

2.5 Constraints

2.5.1 Professor Constraints

Professors can't be scheduled multiple times during one time block

$$\sum_{cik} X_{pcistk} \leq 1 \quad \forall \quad P, S, T, [pcistk] \in \{XV\}, p \notin \text{part-timer}, t \notin \{\text{placeholder}\} \tag{1}$$

Professors can't be scheduled multiple times during intercepting time blocks

$$\sum_{cik} X_{pcistk} + \sum_{cik} X_{pcis,tt,k} \leq 1 \quad \forall \quad P, S, T, TT, [t, tt] \in \{tc\}, [pcistk] \in \{XV\}, [pcis, tt, k] \in \{XV\}, p \notin \text{part-timer} \tag{2}$$

Professors must teach a required number of sections

$$\sum_{cikst} X_{pcistk} = pt_p - ProfTV_p \quad \forall \quad P, [pcistk] \in \{XV\}, p \notin \text{part-timer} \tag{3}$$

Professors can only teach courses they are qualified to teach (removed by XV)

$$\sum_{istk} X_{pcistk} <= pt_p * pq_{pc} \quad \forall \quad P, C, [pcistk] \in \{XV\} \tag{4}$$

Counts the maximum number of days a professor is teaching per each semester

$$\sum_{\substack{citk \\ [t,dw] \in \{TDW\}}} X_{pcistk} <= M * PDayT_{p,s,dw} \quad \forall P, S, DW, [pcistk] \in \{XV\} \tag{5}$$

Counts the number of occurences a professor has back to back classes (in different classrooms)

$$\sum_{i} X_{pcistk} + X_{pci,s,tt,kk} = ProfBBV_{pcist,tt,k,kk} \quad \forall_{---} \text{ where } k \neq kk, [t, tt], \in \{tb\} \tag{6}$$

2.5.2 Time of Day Constraints

Counts the maximum and minimum number of sections offered over different times of day.

$$\sum_{\substack{pcitk \\ [t,td] \in \{TTD\}}} X_{pcistk} \leq TimeMax_s \quad \forall \quad S, TD, [pcistk] \in \{XV\} \tag{7}$$

$$\sum_{\substack{pcitk \\ [t,td] \in \{TTD\}}} X_{pcistk} \geq TimeMin_s \quad \forall \quad S, TD, [pcistk] \in \{XV\} \tag{8}$$

2.5.3 Day of Week Constraints

Counts the maximum and minimum number of sections offered over different days of week. Since 1.5 hour blocks are distributed between 2 days, while 3 hour blocks distributed over 1 day.

$$\sum_{\substack{pcitk,bl \\ [t,dw] \in \{TDW\}, [t,bl] \in \{TBL\}}} X_{pcistk} * tbw_{bl} \leq DayMax_s \quad \forall \quad S, DW, [pcistk] \in \{XV\} \tag{9}$$

$$\sum_{\substack{pcitk,bl \\ [t,dw] \in \{TDW\}, [t,bl] \in \{TBL\}}} X_{pcistk} * tbw_{bl} \geq DayMin_s \quad \forall \quad S, DW, [pcistk] \in \{XV\} \tag{10}$$

2.5.4 Classroom Constraints

Each classroom can only be occupied once per period and the time slots that share periods must also be respected by classrooms.

$$\sum_{pci} X_{pcistk} \leq 1 \quad \forall \quad K, S, T, [pcistk] \in \{XV\}, k \notin \{\text{other}\}, t \notin \{\text{placeholder}\} \tag{11}$$

$$\sum_{pci} X_{pcistk} + \sum_{pci} X_{pcisttk} \leq 1 \quad \forall \quad K, S, T, TT, [t, tt] \in \{tc\}, [pcistk] \in \{XV\}, k \notin \{\text{other}\} \tag{12}$$

Classroom capacities must be respected. The section attendance cannot be higher than the classroom capacity and must be higher than at least 80 percent of the classroom capacity. (removed by XV)

$$\sum_{\substack{pt,td \\ [t,td] \in \{TTD\}}} X_{pcistk} = 0 \quad \forall \quad C, I, S, K, 0.8kc_k \leq csa_{cis} \leq kc_k, [pcistk] \in \{XV\}, k \notin \{\text{other}\} \tag{13}$$

Finally the classroom utilization must be maximized which is done by counting the number of courses allocated to classroom and that value is rewarded in the objective function. (this is not needed as it is modelled by minimizing "other" classrooms)

$$\sum_{pcist} X_{pcistk} = KCount_k \quad \forall \quad K, [pcistk] \in \{XV\}, k \notin \{\text{other}\} \tag{14}$$

2.5.5 Course Constraints

For each course section, there must be scheduled a required number of sections per each time of the day.

$$\sum_{cpstk} X_{pcistk} = 1 \quad \forall \quad I, [pcistk] \in \{XV\} \tag{15}$$

Three hour courses with more than 75 students can only be scheduled at night. (removed by XV)

$$\sum_{pkt, TTD \neq 0, TBL=1} X_{pcistk} = 0 \quad \forall \quad CS_{cis,td} \text{ where } csa_{cis} >= 75, TD \notin \{night\}, BL \in \{3hour\} \tag{16}$$

Some courses are restricted to either 1.5 or 3 hour blocks.

$$\sum_{\substack{pistk \\ [t,bl] \in \{TBL\}}} X_{pcistk} = 0 + CBLPen_{c,bl} \quad \forall \quad C, BL, [pcistk] \in \{XV\}, [c, bl] \in \{cb\} \tag{17}$$

Some courses should not be scheduled on the 3 hour Friday night block.

$$\sum_{\substack{pistk \\ [t,\text{Fri}] \in \{TDW\} \\ [t,Night] \in \{TTD\}}} X_{pcistk} = 0 + CFNPen_c \quad \forall \quad C, [pcistk] \in \{XV\}, c \in \{cf\} \tag{18}$$

2.5.6 Concurrency Constraints

Sections of same courses must not be scheduled together

$$\sum_{\substack{pik \\ p \notin \{\text{part-timer}\}}} X_{pcistk} \leq 1 + ConcI_{cst} \quad \forall \quad C, S, T, [pcistk] \in \{XV\}, t \notin \{\text{placeholder}\} \tag{19}$$

Some concurrent courses must be scheduled together ($[pcistk] \in \{XV\}$ and $[p, cc, istk] \in \{XV\}$)

$$\sum_{\substack{pik \\ p \notin \{\text{part-timer}\}}} X_{pcistk} = \sum_{\substack{pik \\ p \notin \{\text{part-timer}\}}} X_{p,cc,istk} + ConcT_{c,cc,st} \quad \forall \quad C, CC, S, T, t \notin \{\text{placeholder}\}, [c, cc] \in \{cct\} \tag{20}$$

Some concurrent courses cannot be scheduled together ($[pcistk] \in \{XV\}$ and $[p, cc, istk] \in \{XV\}$)

$$\sum_{\substack{pik \\ p \notin \{\text{part-timer}\}}} X_{pcistk} + \sum_{\substack{pik \\ p \notin \{\text{part-timer}\}}} X_{p,cc,istk} \leq 1 + ConcN_{c,cc,st} \quad \forall \quad C, CC, S, T, t \notin \{\text{placeholder}\}, [c, cc] \in \{ccn\} \tag{21}$$

2.6 Objective Function

The model is solved in a hierarchical manner, where each hard constraint is evaluated and then bounded. Then each soft objective is solved and bounded.

a few notes: (1) there is no need to minimize part-timers, because 1st step ensures we provide all possible courses to full-time profs first. (2) since concurrencies and time placeholders are linked, they are minimized together

1. Identify the number of professor teachings violation

$$min \sum ProfTV_p \quad \Rightarrow \quad \sum ProfTV_p \leq \text{ value of } \sum ProfTV_p \tag{22}$$

2. Identify the number of course concurrency violations and placeholder times

$$\begin{aligned} & min \sum (ConcI + ConcT + ConcN) + \sum_{t \in \{\text{placeholder}\}} X \quad \Rightarrow \\ & \sum (ConcI + ConcT + ConcN) \leq \text{ value of } \sum (ConcI + ConcT + ConcN) \\ & \sum_{t \in \{\text{placeholder}\}} X \leq \text{ value of } \sum_{t \in \{\text{placeholder}\}} X \end{aligned} \tag{23}$$

3. Minimize DMS classrooms

$$\min \sum_{k \in \{\text{other}\}} X \Rightarrow \sum_{k \in \{\text{other}\}} X \leq \text{ value of } \sum_{k \in \{\text{other}\}} X \tag{24}$$

4. Final mixed objective

$$\begin{aligned} max \quad & -10 \sum_{s} (TimeMax_s - TimeMin_s) \\ & -10 \sum_{s} (DayMax_s - DayMin_s) \\ & -3 \sum_{c,bl} CBLPen_{c,bl} && [c, bl] \in \{cb\} \\ & -3 \sum_{c} CFNPen_c \\ & +1 \sum_{pcistk} X * pcp_{pc} && [pcistk] \in XV, pcp_{pc} > 0 \\ & +1 \sum_{pcistk,td} X * ptp_{ps,td} && [pcistk] \in XV, [t, td] \in \{TTD\}, ptp_{ps,td} > 0 \\ & +1 \sum_{pcistk,bl} X && [pcistk] \in XV, [t, bl] \in \{TBL\}, pc \in \{pbp\} \\ & -1 \sum_{s,p,dw} PDayT_{p,s,dw} * ppp_p && [t, dw] \in \{TDW\}, p \notin \{\text{Part-Timer}\} \end{aligned} \tag{25}$$